

# Dynamic Recursive Meta-Mathematics Core Specification

## Purpose

This document formalizes the *referee-proof* mathematical center of DRMM as a certifiable, prime-indexed recursion with a strict separation between:

- **Structural layer:** prime indexing, multiplicity counters, PGF objects (bookkeeping; *not* a gain).
- **Analytic layer:**  $\Lambda_m$  as a real scalar gain selected/clamped by CSC to certify contraction.

All correctness results below hold for any finite index set; *primes are used only as a deterministic sparse schedule*.

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## Standing Assumptions (A–D)

**Assumption A (Space + norm).**  $(\mathcal{X}, \|\cdot\|)$  is a Banach space over  $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ . Operator norms are induced:

$$\|A\| := \sup_{\|x\|=1} \|Ax\|, \quad A \in \mathcal{B}(\mathcal{X}).$$

**Assumption B (Forward invariance).** There exists a nonempty closed set  $\Omega \subseteq \mathcal{X}$  such that for all  $t$ ,

$$x \in \Omega \implies \Phi_t(x) \in \Omega,$$

where  $\Phi_t$  is the DRMM update map defined below.

**Assumption C (Bounded sector operator).** The linear sector sum

$$\Xi(t) := \sum_{p \in P_N} a_p(t) U_p(t)$$

(with  $P_N$  a finite index set, often primes) is well-defined in  $\mathcal{B}(\mathcal{X})$  for all  $t$ , and admits the computable bound

$$\|\Xi(t)\| \leq \sum_{p \in P_N} |a_p(t)| \|U_p(t)\|.$$

**Assumption D (Nonlinearity regularity).**  $T : \Omega \rightarrow \mathcal{X}$  is  $L_T$ -Lipschitz on  $\Omega$  (or admits a known upper bound  $L_T$  on  $\Omega$ ):

$$\|T(x) - T(y)\| \leq L_T \|x - y\| \quad \text{for all } x, y \in \Omega.$$

**Prime indexing disclaimer.** Prime indexing is a deterministic sparse schedule; all correctness results hold for any finite index set. Primes are used only as an engineering choice for reproducible sparsity.

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Two quantities with distinct roles:

- **Structural multiplicity functional:**  $\widehat{\Lambda}(t)$ . Generated by multiplicity/PGF bookkeeping; descriptive and type-preserving; **not** used directly as a stability gain.
- **Analytic certification gain:**  $\Lambda_m(t) \in \mathbb{R}_{\geq 0}$ . Selected/clamped by CSC to satisfy explicit contraction inequalities.

**Operational rule:**  $\widehat{\Lambda}(t)$  may propose scale; **CSC clamps** to certified  $\Lambda_m(t)$ .

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## 2) DRMM dynamics (certifiable form)

### 2.1 Sector operator (structural layer)

Let  $P_N = \{p_1, \dots, p_N\}$  be a finite index set (often primes). For each  $t$ :

- $U_p(t) \in \mathcal{B}(\mathcal{X})$  are bounded linear operators (“sectors/channels”).
- $a_p(t) \in \mathbb{K}$  are weights.

Define

$$\forall i(t) := \sum_{p \in P_N} a_p(t) U_p(t) \in \mathcal{B}(\mathcal{X}).$$

### 2.2 Nonlinear map (analytic layer)

Let  $T : \Omega \rightarrow \mathcal{X}$  satisfy Assumption D.

### 2.3 Discrete-time DRMM update

Define the time-varying update map

$$\Phi_t(x) := i(t)x + \Lambda_m(t)T(x), \quad x \in \Omega.$$

Trajectories satisfy

$$X_{t+1} = \Phi_t(X_t).$$


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### 3) Lawfulness budgets (measurable constraints)

These are the auditable constraints that CSC enforces.

Assume there exists  $\varepsilon \in (0, 1)$  such that for all  $t$ :

$$\|\dot{X}(t)\| \leq 1 - \varepsilon.$$

#### (ii) Nonlinear gain budget

Define

$$c := \sup_t \lambda_m(t), L_T.$$

Require  $c < \varepsilon$ .

#### (iii) Optional commutator module (only if used)

If augmenting dynamics with an operator coupling term,

$$\dot{X}_{t+1} = \dot{X}(t)X_t + \lambda_m(t)T(X_t) + \kappa C(X_t),$$

then either:

- **(Energy-preserving option):** show  $C$  is norm-nonincreasing in the chosen geometry (e.g., generated by skew-adjoint flow in a Hilbert norm), or
- **(Budgeted option):** assume  $C$  is Lipschitz on  $\Omega$  with constant  $L_C$ , and fold  $\kappa L_C$  into the gain budget:

$$\sup_t \big(\lambda_m(t)L_T + \kappa L_C\big) < \varepsilon.$$

### 4) Core theorems

#### Theorem 1 (Uniform incremental contraction)

Assume Standing Assumptions (A–D). Suppose there exist  $\varepsilon \in (0, 1)$  and  $c \in [0, \varepsilon)$  such that for all  $t$ :

$$\|\dot{X}(t)\| \leq 1 - \varepsilon, \quad \lambda_m(t), L_T \leq c.$$

Define  $\rho := 1 - \varepsilon + c \in (0, 1)$ . Then for all  $x, y \in \Omega$ :

$$\|\Phi_t(x) - \Phi_t(y)\| \leq \rho^t \|x - y\|,$$

and for any two trajectories  $(X_t), (Y_t)$  driven by the same time-varying maps:

$$\|X_t - Y_t\| \leq \rho^t \|X_0 - Y_0\| \quad (t \geq 0).$$

**Proof sketch (one line):**  $\|\Phi_t(x) - \Phi_t(y)\| \leq (\|\Xi(t)\| + \Lambda_m(t)L_T)\|x - y\| \leq (1 - \varepsilon + c)\|x - y\|.$

### Corollary 1 (Time-invariant fixed point)

If additionally  $\Xi(t) \equiv \Xi$  and  $\Lambda_m(t) \equiv \Lambda_m$  are constant in  $t$ , then  $\Phi(x) = \Xi x + \Lambda_m T(x)$  is a contraction on  $\Omega$ . Hence there exists a **unique** fixed point  $x^* \in \Omega$  and every trajectory with  $X_0 \in \Omega$  converges to  $x^*$ .

### Optional Corollary 2 (Convergent/slowly-varying maps: tracking)

If  $\Phi_t \rightarrow \Phi_\infty$  uniformly on  $\Omega$  and all  $\Phi_t$  share the same contraction modulus  $\rho < 1$ , then  $(X_t)$  tracks the unique fixed point of  $\Phi_\infty$  (standard contracting time-varying system tracking lemma).

## 5) Forward-invariance: a constructible sufficient condition

### Lemma (Ball invariance via a growth bound)

Let  $\Omega = B_R := \{x \in \mathcal{X} : \|x\| \leq R\}$ . Suppose that for all  $\|x\| \leq R$  and all  $t$ :

$$\|\Xi(t)x\| \leq \|\Xi(t)\| R, \quad \|T(x)\| \leq \beta + L_T \|x\|,$$

for some  $\beta \geq 0$ . If for all  $t$ :

$$\|\Xi(t)\| R + \Lambda_m(t)(\beta + L_T R) \leq R,$$

then  $B_R$  is forward-invariant:  $x \in B_R \Rightarrow \Phi_t(x) \in B_R$ .

## 6) CSC: Certified Spectral Control (auditable per-step certification)

### 6.1 Certificate quantities

Given computable bounds  $\widehat{\|\Xi(t)\|}$  and  $\widehat{L_T(t)}$  (global or local), define

$$\widehat{\rho}(t) := \widehat{\|\Xi(t)\|} + \Lambda_m(t) \widehat{L_T(t)},$$

$$\widehat{\varepsilon}(t) := 1 - \widehat{\|\Xi(t)\|}, \quad \widehat{c}(t) := \Lambda_m(t) \widehat{L_T(t)}.$$

Return the per-step certificate tuple

$$\mathrm{CSC}(t) := (\widehat{\varepsilon}(t), \widehat{c}(t), \widehat{\rho}(t)),$$

with pass/fail criterion  $\hat{\rho}(t) < 1$ .

Define

$$\Lambda_{\mathrm{target}}(t) := \max\left\{0, \frac{\widehat{\varepsilon}(t) - \delta}{\widehat{L}_T(t)}\right\}.$$

The certified gain is clamped as

$$\Lambda_m(t) := \min\{\Lambda_{\max}, \Lambda_{\mathrm{target}}(t), \widehat{\Lambda}(t)\}.$$

### 6.3 Global / local / hybrid policy

- **Global:** use a conservative  $L_T^{\mathrm{glob}}$  for  $\widehat{L}_T(t)$ .
- **Local:** use online estimates (Jacobian norm, secant slopes, Lipschitz trackers).
- **Hybrid (recommended):** use local estimates when reliable; otherwise fall back to global.

## 7) CSC pseudocode (plug-and-play)

Algorithm CSC(t): Per-step certificate + gain clamping

Inputs:

- Sector set  $P_N$ ; operators  $U_p(t)$ ; weights  $a_p(t)$
- Proposal  $\widehat{\Lambda}(t)$  from structural PGF/multiplicity layer
- Safety margin  $\delta > 0$ ; gain cap  $\Lambda_{\max} > 0$
- Global Lipschitz bound  $L_T^{\mathrm{glob}}$  (optional)
- Local estimator for  $\widehat{L}_T(t)$  (optional)

1) Linear bound:

$\widehat{\Xi}(t) \leftarrow \sum_{p \in P_N} |a_p(t)| \cdot \|U_p(t)\|$  (or tighter computable bound)

$$\widehat{\varepsilon}(t) \leftarrow 1 - \widehat{\Xi}(t)$$

2) Lipschitz bound:

if reliable local estimate available:

$$\widehat{L}_T(t) \leftarrow \text{local estimate}$$

else:

$$\widehat{L}_T(t) \leftarrow L_T^{\mathrm{glob}}$$

3) Clamp target:

$$\Lambda_{\mathrm{target}}(t) \leftarrow \max\{0, (\widehat{\varepsilon}(t) - \delta) / \widehat{L}_T(t)\}$$

4) Clamp:

$$\Lambda_m(t) \leftarrow \min\{\Lambda_{\max}, \Lambda_{\mathrm{target}}(t), \widehat{\Lambda}(t)\}$$

5) Certificate:

$$\begin{aligned}\hat{c}(t) &\leftarrow \Lambda_m(t) \cdot \hat{L}_T(t) \\ \hat{\rho}(t) &\leftarrow \hat{\|\Xi(t)\|} + \hat{c}(t) \\ \text{CSC}(t) &\leftarrow (\hat{\epsilon}(t), \hat{c}(t), \hat{\rho}(t))\end{aligned}$$

6) Audit log (required):

Record  $\hat{\|\Xi(t)\|}$  (and bound method),  $\hat{L}_T(t)$  (and method),  $\hat{\Lambda}(t)$ ,  $\Lambda_m(t)$ , and  $\text{CSC}(t)$ , plus pass/fail  $[\hat{\rho}(t) < 1]$ .

Output:

- Certified gain  $\Lambda_m(t)$
- Certificate  $\text{CSC}(t) = (\hat{\epsilon}(t), \hat{c}(t), \hat{\rho}(t))$

**Application note:** If  $\hat{\rho}(t) < 1$  holds for all  $t$  (or  $\sup_t \hat{\rho}(t) < 1$ ), then Theorem 1 applies with  $\rho := \sup_t \hat{\rho}(t)$  (or any upper bound), yielding uniform incremental contraction on  $\Omega$ .

## 8) Experimental predictions (now fully testable)

1. **Certificate-robust stability:** When CSC reports  $\hat{\rho}(t) < 1$ , observed trajectory distances should contract at rate consistent with  $\rho$  (or its bound).
2. **Prime vs sparse ablation:** Primes should outperform alternatives only insofar as they improve  $\widehat{\|\Xi(t)\|}$  or reduce effective  $\widehat{L}_T(t)$ . Otherwise “prime” behaves like “sparse schedule.”
3. **Commutator benefit (conditional):** Only include commutator modules after verifying skew-adjointness (energy-preserving) or enforcing a commutator Lipschitz budget.

## 9) External references (selected)

The items below provide standard foundations and adjacent tools for the DRMM core results (contraction/ISS/small-gain, incremental stability, induced norms, and practical Lipschitz certification).

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